

Name: Solutions

Student ID: _____

1. (10 points) Expand the following logarithmic functions as much as possible:

(a) $\ln \left(\frac{x^3 \sqrt{y+1}}{(2y)^2} \right)$

(b) $\ln \left(\frac{(x^2 + 3y)^4}{e^{2x} y^3} \right)$

Bonus (+1 point): Take the derivative of (a) w.r.t. $\ln(x)$

$$\begin{aligned}
 (a) \quad & \ln(x^3 (y+1)^{1/2}) - \ln((2y)^2) \\
 &= \ln(x^3) + \ln((y+1)^{1/2}) - (\ln(4) + \ln(y^2)) \\
 &= 3\ln(x) + \frac{1}{2}\ln(y+1) - \ln(4) - 2\ln(y)
 \end{aligned}$$

$$\frac{df}{d\ln(x)} = 3$$

$$\begin{aligned}
 (b) \quad & \ln((x^2 + 3y)^4) - \ln(e^{2x} y^3) \\
 &= 4\ln(x^2 + 3y) - \ln(e^{2x}) - \ln(y^3) \\
 &= 4\ln(x^2 + 3y) - 2x\ln(e) - 3\ln(y) \\
 &= 4\ln(x^2 + 3y) - 2x - 3\ln(y)
 \end{aligned}$$

2. (10 points) Kelsey has utility function $U(x, y) = x^{1/4}y^{3/4}$, and she faces prices $p_x = 1$ and $p_y = 2$. How much income does she need to achieve utility level $U = 8$?

(Helpful calculations: $2^{3/4} \approx 1.68$; $3^{3/4} \approx 2.28$. But you may leave your answer un-simplified)

$$x^m = \frac{I}{4p_x} \quad y^m = \frac{3I}{4p_y}$$

$$V = \bar{U} = 8 \Rightarrow \left(\frac{I}{4p_x}\right)^{1/4} \left(\frac{3I}{4p_y}\right)^{3/4} = 8$$

$$\frac{I}{4} \left(\frac{1}{p_x}\right)^{1/4} \left(\frac{3}{p_y}\right)^{3/4} = 8$$

$$I \left(\frac{1}{1}\right)^{1/4} \left(\frac{3}{2}\right)^{3/4} = 32$$

$$I = \frac{32}{\left(\frac{3}{2}\right)^{3/4}} \approx 23.6$$

$$V = \left(\frac{I}{4}\right)^{1/4} \left(\frac{3I}{8}\right)^{3/4} = 8$$

$$I = \frac{8}{\left(\frac{1}{4}\right)^{1/4} \left(\frac{3}{8}\right)^{3/4}} \approx 23.6$$

Bonus: Who is the current Chair of the UCLA Economics Department?

Kathleen McGarry